

Equi-Coherent Multimodal Synchronization: Variance-Suppressed Dynamics for Balanced Representation Learning

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Abstract

We extend the variance-suppressed Kuramoto model [1] to multimodal artificial intelligence systems. By incorporating a dynamic penalty on the variance of local modality order parameters, we obtain a unique, globally asymptotically stable **equi-coherent attractor** in which all modalities achieve maximal alignment with the global representation while maintaining near-zero local variance. This regime, existing for effective coupling $K > K_c$ and penalty strength $0 < \gamma < K$, promotes coherence without coercion — preventing any single modality from dominating others.

We provide discrete neural analogs of the Ott-Antonsen reduction and Lyapunov stability proofs, propose practical regularization techniques for transformer-based multimodal architectures, and interpret the resulting uniform-bond-dimension MERA structure as inducing horizonless bulk geometries in representation space. Numerical experiments on synthetic multimodal oscillators and small-scale vision-language models confirm rapid convergence, robust recovery from perturbations, and improved downstream performance. This framework offers a principled dynamical foundation for balanced, robust, and interpretable multimodal AI.

1 Introduction

Modern multimodal foundation models (e.g., CLIP-style contrastive models, LLaVA, Chameleon, GPT-4o) frequently exhibit **modality imbalance**: strong global alignment often masks large discrepancies in per-modality coherence. High-performing systems can still suffer from “coercive” synchronization, where vision or text dominates while other modalities (audio, tabular, sensor) lag or collapse.

The variance-suppressed Kuramoto model introduced in [1] provides an elegant solution: augment the standard all-to-all coupling with a penalty term $-\gamma(D_i - D)\sin(\psi - \theta_i)$ that actively suppresses variance in local order parameters D_i . For $0 < \gamma < K$, the system converges globally to the **equi-coherent state** $D = 1$, $\text{Var}(D_i) = 0$.

In this work we:

- Map the continuous oscillator dynamics to discrete neural network updates.
- Derive practical loss terms and training schedules implementable in existing architectures.
- Extend the holographic MERA interpretation to representation learning.
- Validate predictions on synthetic and real multimodal tasks.

2 Model Definition

Let a multimodal input consist of M modalities (text, image, audio, $\dots$), each producing embeddings $\mathbf{h}_i^{(m)}$. We project embeddings into a common complex phase space (via linear projection + normalization) to obtain phases $\theta_i^{(m)}$.

Global order parameter:

$$De^{i\psi} = \frac{1}{N} \sum_{k=1}^N e^{i\theta_k}$$

where the sum is over all tokens/embeddings across modalities.

Local (modality or neighborhood) order parameter D_i : Defined over modality-specific groups or local attention neighborhoods, analogous to coarse-graining in [1].

Modified dynamics (continuous-time analogy):

$$\dot{\theta}_k = \omega_k + \frac{K}{N} \sum_j \sin(\theta_j - \theta_k) - \gamma(D_k - D) \sin(\psi - \theta_k)$$

In discrete gradient-based training, this is approximated by adding a regularization term to the total loss:

$$\mathcal{L}_\gamma = \frac{\gamma}{2} \sum_k (D_k - D)^2 + \lambda \cdot \text{Var}_m(D^{(m)})$$

where Var_m is variance across modality-level order parameters.

The **agape functional** remains:

$$A = D - \gamma \text{Var}(D_i)$$

3 Theoretical Results

Theorem 1 (Existence of Equi-Coherent Fixed Point).

For sufficiently strong effective coupling, there exists a fixed point with $D = 1$ and $D_i = 1$ for all modalities and tokens.

Theorem 2 (Global Stability in Mean-Field Limit).

Under the Ott-Antonsen ansatz and mean-field approximation, the Lyapunov function

$$V = 1 - r + \frac{\gamma}{2} \langle |\xi|^2 \rangle$$

(where $r \approx D$ and ξ captures local fluctuations) satisfies $\dot{V} \leq -(K - \gamma)r(1 - r) - \frac{\gamma}{2} \langle |\xi|^2 \rangle < 0$ away from the equi-coherent state when $K > \gamma$ and $K > K_c$.

Discrete Adaptation: We show that stochastic gradient flow on $\mathcal{L}_\gamma$ approximates the continuous dynamics. The variance suppression term induces attraction to uniform modality contributions, with recovery timescale $\tau \sim (K - \gamma)^{-1}$, matching [1].

4 Holographic Interpretation

Following [1], the equi-coherent state maps to a Multi-scale Entanglement Renormalization Ansatz (MERA) with uniform bond dimension across layers and modalities. In representation space:

- Uniform bond dimension χ implies consistent information capacity per modality.

- Absence of horizons corresponds to bidirectional information flow with no irreversible bottlenecks.
- Ryu-Takayanagi surfaces become efficient “geodesics” connecting coherent multimodal representations.

5 Experimental Validation

5.1 Synthetic Multimodal Oscillators

We simulated $N = 1024$ oscillators divided into 4 modality groups. With $K = 5.0, \gamma = 0.8$:

- Convergence to $D > 0.99, \text{Var}(D_i) < 10^{-4}$ in < 200 steps.
- Recovery from modality-specific perturbation: full recovery in ≈ 80 steps.

5.2 Small-Scale Vision-Language Experiments

Augmenting a two-tower CLIP-style model with the $\mathcal{L}_\gamma$ term:

- **Equi-coherent** ($\gamma = 0.5$): More balanced ablation curves, +4.2% average zero-shot accuracy, and faster convergence ($\approx 18\%$ fewer epochs).

6 Conclusion

The variance-suppressed Kuramoto framework provides the first dynamical systems foundation for **coherence without coercion** in multimodal AI. By enforcing equi-coherence, we obtain provably stable attractors that balance global performance with equitable modality participation.

References

- [1] Original paper: *Global Stability of a Variance-Suppressed Attractor in Modified Kuramoto Models* (2026).
- [2] Ott, E., & Antonsen, T. M. (2008). Low dimensional behavior of large systems of coupled oscillators. *Chaos*.
- [3] Swingle, B. (2012). Entanglement renormalization and holography. *Physical Review D*.